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# A LAB REPORT ON KIRCHHOFF'S LAWS OR RULES

## 1 Theory

Building upon our previous analysis—where a plumbing system was used as an analogy for electrical flow—this report evaluates the validity and application of Kirchhoff's Rules in complex circuits that cannot be simplified through basic series or parallel configurations. Kirchhoff's Rules provide the mathematical framework necessary to solve for unknown variables in multi-loop circuits through two fundamental principles:

### I. Kirchhoff's Voltage Law (KVL)

KVL is a reflection of the Law of Conservation of Energy. It states that the algebraic sum of the potential differences (voltages) around any closed loop or “conductive path” must equal zero. Much like a “summation of forces” in mechanical equilibrium, KVL ensures that the energy supplied by a source is exactly accounted for by the potential drops across the loads.

Mathematically, this is expressed as:

$$\sum_{k=1}^n V_k = 0$$

where:

- $n$  represents the total number of electrical components encountered along a specific closed path.
- $k$  serves as the index for each individual element, effectively “counting” our progress as we traverse the loop.

- $V_k$  is the potential difference (measured in Volts) across the  $k^{th}$  element; by convention, a voltage source (battery) represents a positive gain, while a resistor represents a negative drop.

## II. Kirchhoff's Current Law (KCL)

KCL is a reflection of the Law of Conservation of Charge. It dictates that charge cannot “pile up” or vanish at a junction (node). While our previous analogy compared current to water pressure, KCL specifically describes the flow: the total current entering a junction must exactly equal the total current exiting it.

At any junction, the sum of currents is zero:

$$I_{in} = I_{out_1} + I_{out_2} + \cdots + I_{out_n}$$

where:

- $I$  represents the electric current, measured in Amperes ( $A$ ).
- $n$  represents the number of outgoing branches from the junction.

To test the efficacy of these models, our team analyzed two circuits of varying complexity. This study serves to validate the accuracy of Kirchhoff's Rules and to exercise advanced circuit analysis techniques, providing a foundation for the study of more sophisticated electrical systems.

## 2 Conclusion

### Activity 1: First Circuit

In Activity 1 our team used multimeters to measure the experimental currents and voltage drops across three resistors ( $2220\ \Omega$ ,  $1494\ \Omega$ , and  $820\ \Omega$ ) and compared them to theoretical

values derived from a system of linear equations. Our measurements yielded mixed results; for the  $2220\ \Omega$  and  $1494\ \Omega$  resistors, the measured currents of  $0.0016\ \text{A}$  and  $0.0012\ \text{A}$  exhibited percent errors of  $12.57\%$  and  $7.460\%$  from the expected values of  $0.0018\ \text{A}$  and  $0.0013\ \text{A}$ , respectively. The  $820\ \Omega$  resistor presented a significant anomaly, with a measured voltage drop of  $1.48\ \text{V}$  deviating from the expected  $0.4373\ \text{V}$  by  $238.4\%$ . These discrepancies, particularly the extreme error in the third resistor, suggest that while the circuit followed a recognizable pattern, the results were heavily impacted by systematic errors such as internal meter resistance or contact issues in the breadboard assembly.

## Activity 2: Second Circuit

For Activity 2, our team calculated the expected theoretical currents across each node. The experimental data demonstrated varying levels of precision; we observed relatively accurate current measurements for the  $2220\ \Omega$  and  $1800\ \Omega$  resistors, recording values of  $0.00003070\ \text{A}$  and  $0.0008\ \text{A}$ , which yielded low percent errors of  $0.1478\%$  and  $1.113\%$  compared to theoretical calculations. Conversely, readings for the  $820\ \Omega$ ,  $1494\ \Omega$ , and  $1200\ \Omega$  resistors deviated more significantly, with current errors of  $58.37\%$ ,  $27.28\%$ , and  $23.83\%$  respectively. Despite these fluctuations, the general pattern of the data broadly aligns with the theoretical framework of Kirchhoff's rules. The larger discrepancies are likely attributable to procedural and systematic inaccuracies during data collection—such as inconsistent probe contact or minor assembly errors—rather than a flaw in the underlying physical laws.

## 3 Questions

### Activity 1 Questions

1. **Compare your results in step #2 and #3. Discuss.**

After analyzing and comparing our results to the analog model introduced in the previous laboratory session, we observed that the values at each node are influenced

by preceding circuit elements. This supports the conclusion that current flows in a “river-like” pattern; however, it is important to note that this macro-scale pattern does not necessarily reflect electron behavior at the microscopic level. Instead, distinct physical mechanisms govern charge transport to allow this observed behavior to occur.

## 4 Above and Beyond

While these labs have allowed us to experience firsthand how to understand the changes in voltage and current across a circuit, it was not until we commenced research for this report that we discovered that the flow of energy across wires has less to do with the actual physical displacement of electrons and more to do with the electric fields they produce. It is fascinating how these mechanisms produce this predictable behavior at the macroscopic level. However, further research also provided insight into the limitations that occur in extreme cases—such as why certain wires require specific insulation for protection or the challenges of long-distance intercommunication. While the majority of circuit analysis applications can be successfully modeled using Kirchhoff’s rules, advanced applications where accuracy and timing are fundamental require us to extend our models to account for these underlying physical complexities.